Assignment 2: Model Stealing Detection

Anonymous authors
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Varun Chaturvedi (7087955),Vaishnavi Anil Borse (7075846), Team 50

1 Assignment

In this task we are given a white-box access to a target ResNet-18 image classifier which is trained on a subset of CIFAR-100 and 360 suspect models. Our goal is to determine which of the suspect models were stolen or derived from the target model. This is done by assigning a continuous stealing confidence score to each. A model is considered stolen if it was directly copied, fine-tuned, distilled, or derived from the target model or any of its intermediate checkpoints. Even if post-processing or further training was applied afterward. The evaluation metric is TPR at 5% FPR, which rewards the methods that correctly flag stolen models while maintaining a low false positive rate.

2 Methodology

Our approach compares the functional behaviour of each suspect model against the target model by running both on the same fixed set of inputs (the CIFAR-100 test set). And measures the similarity of their output logits. All experiments used 40 batches of 256 images (10,240 samples total). Code is available at: `https://github.com/arcnova2095/TML-Model-Stealing`

2.1 Cosine Similarity on Raw Logits

Our baseline computes the cosine similarity between the flattened logit vectors of the target and each of the suspect model:

$$\text{score} = \frac{\mathbf{t} \cdot \mathbf{s}}{\|\mathbf{t}\| \, \|\mathbf{s}\|} \tag{1}$$

where $\mathbf{t}$ and $\mathbf{s}$ are the concatenated logit vectors over all evaluation samples. This resulted a score of **0.537037**. The variants using softmax-normalised logits and log-softmax-normalised logits were also tested but did not improve over the raw cosine baseline.

2.2 KL Divergence

Next we replaced the cosine similarity with the KL divergence between the softmax output distributions of the target and suspect and evaluated per sample and averaged across the dataset. To obtain a confidence score (higher = more stolen), we converted the divergence to a similarity via:

$$\text{score} = \frac{1}{1 + D_{\mathrm{KL}}(p_{\text{suspect}} \,\|\, p_{\text{target}})} \tag{2}$$

Softmax temperature was set to $T = 1$ to soften the distributions. This method improved performance to **0.555556**, making it our best method.

2.3 Ensemble of Cosine Similarity and KL Divergence

We explored linear combinations of the two scores:

$$\text{score}_{\text{ens}} = \alpha \cdot \text{score}_{\text{cos}} + (1 - \alpha) \cdot \text{score}_{\text{KL}} \tag{3}$$

for $\alpha \in \{0.5, 0.4, 0.3, 0.2, 0.1, 0.05\}$. None of the ensemble configurations improved over the standalone KL divergence score, suggesting that cosine similarity does not add complementary signal in this setting.

2.4 Summary of Results

Table 1: Scores (TPR @ 5% FPR) for all methods evaluated.

Method	**Score**
Cosine Similarity (raw logits)	0.537037
KL Divergence ($T = 1$)	**0.555556**

The best result on the public leaderboard was **0.555556**, achieved with the KL divergence method. Ensembling and Cosine Similarity did not yield any improvement.

3 Conclusion

Our results showed that comparing output distributions via KL divergence is more informative than the cosine similarity for stolen model detection. Since KL divergence captures the asymmetric distributional differences that cosine similarity ignores. The implications of model stealing are significant: stolen models allow adversaries to replicate expensive proprietary systems at negligible cost, and to use them for malicious purposes such as crafting adversarial examples or violating intellectual property rights. Reliable detection of stolen models is therefore a key building block for trustworthy machine learning ecosystems.